

m_{phi}	Coherent photon background					DIS background				
	Paper [pb]	Our estimate +/- MC [pb]	Ratio ours/paper	Difference [%]	Within 10%?	Paper [pb]	Our estimate +/- MC [pb]	Ratio ours/paper	Difference [%]	Within 10%?
0.010	0.58	2.2707 +/- 0.089	3.9149	291.49	No	0.031	0.03233 +/- 0.00066	1.0430	4.30	Yes
0.032	0.58	2.2707 +/- 0.089	3.9149	291.49	No	0.031	0.03233 +/- 0.00066	1.0430	4.30	Yes
0.100	0.58	2.2707 +/- 0.089	3.9149	291.49	No	0.031	0.03233 +/- 0.00066	1.0430	4.30	Yes
0.316	0.58	2.2707 +/- 0.089	3.9149	291.49	No	0.031	0.03233 +/- 0.00066	1.0430	4.30	Yes
1.000	0.53	1.9376 +/- 0.011	3.6558	265.58	No	0.052	0.05454 +/- 0.00085	1.0489	4.89	Yes
1.585	0.53	1.9278 +/- 0.011	3.6373	263.73	No	0.052	0.05454 +/- 0.00085	1.0489	4.89	Yes
2.512	0.41	1.4969 +/- 0.008	3.6511	265.11	No	0.052	0.05450 +/- 0.00085	1.0481	4.81	Yes
3.981	0.19	0.6955 +/- 0.0056	3.6604	266.04	No	0.033	0.03489 +/- 0.00068	1.0573	5.73	Yes
5.000	0.11	0.4252 +/- 0.0038	3.8652	286.52	No	0.0204	0.02134 +/- 0.00054	1.0459	4.59	Yes
6.310	0.081	0.3240 +/- 0.0048	3.9996	299.96	No	0.0155	0.01647 +/- 0.00047	1.0623	6.23	Yes
10.000	0.081	0.3327 +/- 0.0064	4.1070	310.70	No	0.0155	0.01637 +/- 0.00047	1.0561	5.61	Yes

Table 1. Completed independent cross-check of the effective coherent-photon and DIS background cross sections reported in Table I of arXiv:2505.08871v3. The published values are compared with our calculations after applying the same mass-dependent cuts. For the coherent-photon process $e \text{ Au} \rightarrow e \text{ Au} \gamma$, we used the coherent matrix element with the Helm nuclear form factor, 18 GeV electrons, a 100 GeV-per-nucleon gold beam, and a 10^{-6} GeV photon-mass regulator. The detector-miss probability was included directly in the TFOAM integration: 10^{-6} for $|\eta_{\gamma}| \leq 3.5$ and 1 outside this range. Three independent seeds were combined, and the quoted MC uncertainty is the larger of the seed standard error and the propagated integration error. For DIS, we generated 100,000 unweighted leading-order $e p \rightarrow e j$ events with MG5_aMC@NLO 3.7.0 and CT10nlo, applied the Table I electron cuts, and multiplied the selected rate by $A=197$, the 5% veto-survival probability, and the 10^{-6} jet-miss probability. **Conclusion:** every DIS point agrees with the published estimate within 4.30-6.23%, whereas our photon result is 3.64-4.11 times the published value and fails the 10% criterion at every mass. This establishes a robust DIS cross-check, while the photon normalization still requires comparison with the authors' corrected calculation and unpublished integration settings. Difference [%] = $100 \times (\text{our value} - \text{paper value}) / \text{paper value}$; all uncertainties shown are Monte Carlo uncertainties only.